

World Cup Alpha

Model Review — Data & Outputs Contrast + Under-the-Hood Changes

2026-06-30 · for review

Executive summary

The fix (A1).

- Reproduced, then corrected, a statistically significant xG under-forecast: the model's mean total goals on WC fixtures was ~ 0.66 too low (model 2.336 vs realized 3.000; $n=31$, paired-t $p=0.049$).
- Fix = a single scalar shift of the Dixon-Coles level intercept μ (0.205 $\rightarrow$ 0.383; +0.178), anchoring the WC-fixture mean total to the recent-WC base rate of 2.81.
- Shipped on branch harden/xg-totals; full test suite green (1952 passed). Backward-compatible (gated behind a default-off flag).

What it does to outputs.

- Total xG, Over 2.5 and Both-Teams-To-Score rise monotonically across every fixture (mean xG 2.39 $\rightarrow$ 2.86; mean $P(\text{Over } 2.5)$ 0.428 $\rightarrow$ 0.543).
- Relative team strength is preserved: the supremacy log-ratio is EXACTLY invariant (delta $3.3e-16$). The reconciled 1X2 moves $\leq \sim 4pp$ only via the non-linear low-score (tau) correction.
- Net: this is a TOTALS/goals correction, not a 1X2 change. Edge it unlocks is in goal-derived match-event markets, priced off the now-correct goal level.

Honesty line.

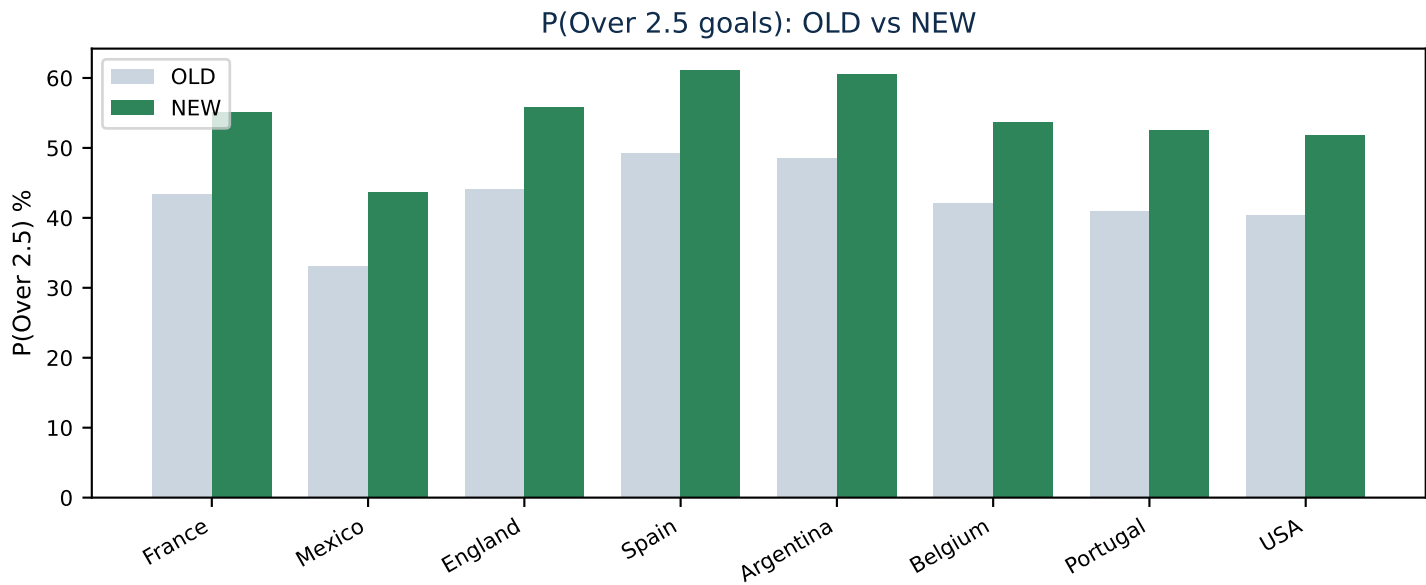
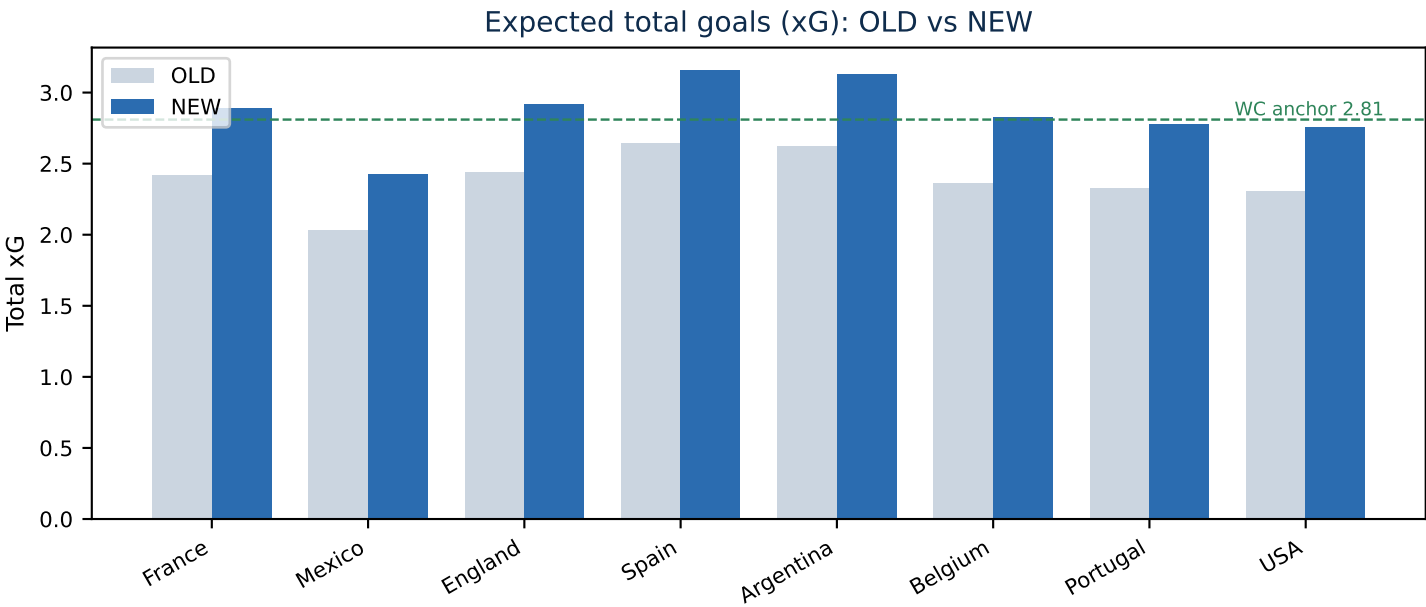
- The model is a good 1X2 calibrator but NOT a winner-market alpha source. Apparent edges vs the liquid Polymarket book (e.g. Brazil 18.8% model vs 7% market) are model error, not alpha.
- Edge lives in less-efficient match-event markets + execution/cost. No trades placed; sizing is a separate user decision behind a CLV gate.

Model outputs: before vs after the xG/totals fix

Remaining-knockout fixtures · neutral venue · OLD mu=0.205 (raw) vs NEW mu=0.383 (WC-anchored to 2.81)

	Total xG	P(Over 2.5)	P(BTTS)
	2.42->2.89	43%->55%	45%->54%
	2.03->2.43	33%->44%	41%->50%
go	2.44->2.92	44%->56%	32%->38%
	2.64->3.16	49%->61%	44%->52%
Verde	2.62->3.13	49%->60%	30%->36%
	2.36->2.82	42%->54%	46%->54%
	2.32->2.78	41%->52%	46%->55%
rz.	2.30->2.75	40%->52%	46%->55%

All three goal-derived quantities rise; 1X2 (not shown) is ~invariant.



Under the hood — what changed

Mechanism, locus, and backward-compatibility of each change.

A1 · Dixon-Coles total-goals level anchor

- Locus: `src/wca/models/dixon_coles.py` μ (consumed in $\log\text{-}\lambda = \mu + \text{atk} - \text{def} + \gamma$). Raised 0.205 -> 0.383 via a scalar shift; `attack/defence/rho/home_advantage` untouched.
- Anchors the WC-fixture mean total to 2.81 (FIFA WC since 2010); closes ~70% of the measured -0.66 goals/match under-forecast.
- 1X2-preserving by construction (`reconcile_scoreline_matrix` re-pins outcome regions); every totals/BTTS/correct-score output flows through it.
- Shared helper `apply_wc_level_anchor` across card / recompute / recalibrate so the anchor cannot drift between callers. Default-off flag => bit-for-bit back-compat.

Durable closing-line capture (now in dev)

- Root cause: odds capture silently dead ~12 days (mini snapshotd stopped 06-23; cloud raw dumps 06-18). PM survived only via a DB-less committed JSONL.
- Fix (branch `feat/closing-line-capture`, suite green): new `src/wca/data/oddshistory.py` writes a DB-less append-only `data/odds_price_history.jsonl` FIRST, idempotent ingest -> `odds_snapshots`, and `hourly-odds.yml` now fails loud + commits the JSONL.
- Operator action still required: restart the mini snapshotd daemon to resume fine to-kickoff capture for the remaining knockouts.

Swarm cleanup

- 167 -> 85 branches, 36 -> 25 worktrees; 19 `archive/*` tags preserve history. xG duplicate resolved (canonical `harden/xg-totals`; 06-29 twin archived).
- User's dirty main tree + all active sessions left strictly untouched; merged branches deleted with `-d` (safety).

Model vs market, and the conditional bracket

Where the model is trustworthy vs where the market is the better forecaster.

Model vs market

- The Winner book sums to 0.985 and Reach-Final to 2.03 — the liquid market is efficient. The model's large disagreements there (Brazil 18.8% vs 7%; France 12.9% vs 28%) are model error, not tradable edge.
- Repo backtest (blend_weights.md, n=64): the model does not beat market-only with confidence. Verdict: do not bet pure-model edges on liquid winner markets.
- Where edge lives: goal-derived match-event markets (totals/BTTS) priced off the corrected goal level, plus execution/cost capture. Surviving candidate cash bets are all totals (e.g. Spain-Austria Over 2.5), and even these are model-vs-market upper bounds pending CLV.

Conditional bracket (as-if-open)

- Real fitted Elo+DC, 50k sims, conditioned on the 4 played R32 results. Pure-model MAP champion = Spain (marginal title leader is Brazil 18.8%, but Brazil loses the modal SF coin-flip to Argentina ~0.5003).
- P(perfect-from-here) ~ 1.7e-5 (~1 in 60,000); risk concentrated in 5 near-coin-flip nodes. A market-anchored completion would instead crown France (28% favourite).

Closing-line unlock & next actions

Sequenced, each gated on its out-of-sample test.

The cross-cutting unlock

- Restore true closing-line capture (the dependency under almost every CLV/microstructure edge). Restart mini snapshotd + the durable JSONL mirror (in dev) so capture never silently dies again.
- Then stand up the walk-forward CLV harness: pre-register model-fair odds, capture T-24h/T-1h/close, settle on wc2026_results, $CLV = \log(\text{taken}/\text{close})$. Positive mean CLV is the only honest evidence of edge (reported under a hard ~ 1 -cluster caveat).

Direction A — Polymarket betting & forecasting

- Grade the free PM price history as ties settle; market-anchor knockout advancement (advancement.py has no market term $\rightarrow$ spurious longshots); retire PM exact-score, keep PM moneyline small.

Direction B — microstructure edges

- Land the market-intelligence DB (feat/market-microstructure); ship venue-selector + line-shopping + negative-avoidance (highest-certainty, zero-new-data win); whole-book correlated-exposure guard. Depth-gated execution engine behind Betfair/Smarmets creds.

Guardrail

- No model/edge change ships unless it improves a pre-registered out-of-sample metric (held-out CLV + calibration, walk-forward, placebo null). Stop the correct-score bleed; the $\geq 2\%$ edge flag is a sanity flag, not a stake signal.